Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology (B.Sc.I.T Semester III) Python For Data Science Practical Practical – V

Roll No.: S021

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Class:SYBsc IT

Subject: DATA STRUCTURE

Date of Assignment:12/08/2026

Date/Time of Submission:12/08/2026 **bold text**

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame objects. bold text

```
import pandas as pd

# Create a DataFrame using a dictionary
data = {
    "Name": ["DHARMI", "DIVYESH", "SMIT", "HEMISH", "URVISH"],
    "Age": [20, 21, 19, 22, 34 ],
    "Marks": [85, 90, 78, 45, 67]
}

df = pd.DataFrame(data)

print(df)
print("IQRA S021")
```

```
      Name  Age  Marks
0   DHARMI   20     85
1  DIVYESH   21     90
2     SMIT   19     78
3   HEMISH   22     45
4   URVISH   34     67
IQRA S021
```

5b. Write a python code to show statistical information on a given data set. MEAN **bold text**

```
import statistics as st

# Calculate average values
print(st.mean([1, 3, 5, 7, 9, 11, 13]))
print(st.mean([1, 3, 5, 7, 9, 11]))
print(st.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("IQRA S021")
```

```
7
6
1.8666666666666667
IQRA S021
```

MEDIAN:**bold text**

```
import statistics as st

# Calculate middle values
print(st.median([1, 3, 5, 7, 6, 34, 13]))
print(st.median([1, 3, 5, 7, 9, 11]))
print(st.median([-11, 5.5, -3.97, 7.1, -9, 78]))
print("IQRA S021")
```

```
6
6.0
0.7649999999999999
IQRA S021
```

MODE:

```
import statistics as st

# Calculate the mode
print(st.mode([1, 3, 3, 3, 5, 7, 7 ,9, 11]))
print(st.mode([1, 1, 3, -5, 7, -9, 11]))
print(st.mode(['blue', 'purple', 'green', 'red']))
print("IQRA S021")
```

```
3
1
blue
IQRA S021
```

VARIANCE

```
import statistics as st

# Calculate the variance from a sample of data
print(st.variance([1, 3, 5, 7, 9, 11]))
print(st.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(st.variance([-11, 5.5, -3.4, 7.1]))
print(st.variance([1, 30, 50, 100]))
print("IQRA S021")
```

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
IQRA S021
```

STANDARD **DEVIATION**

```
import statistics as st

# Calculate the standard deviation from a sample of data
print(st.stdev([1, 3, 5, 7, 9, 11]))
print(st.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(st.stdev([-11, 5.5, -3.4, 7.1]))
print(st.stdev([1, 30, 50, 100]))
print("IQRA S021")
```

```
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007
IQRA S021
```

5c.Write a python code to create a pandas series from dictionaries.

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'A': 10, 'B': 20, 'C': 30}

# create a series
series = pd.Series(dictionary)

print(series)
print("IQRA S021")
```

```
A    10
B    20
C    30
dtype: int64
IQRA S021
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays. bold text

```
# importing pandas as pd
import pandas as pd
```

```
import pandas as pd

# dictionary of lists
dict = {
    'name': ["DHARMI", "SMIT", "HEMISH", "DIVYESH", "URVISH"]
    ,
    'degree': ["ENGINEERING", "MECHANICAL", "DATA SCIENCE", "BCA", "MBBS"],

    'hobbies': ["dancing", "singing", "gaming", "sports", "swimming"]

}

df = pd.DataFrame(dict, index=[True, False, True, False, True])

print(df)

print("IQRA S021")
```

```
          name        degree   hobbies
True    DHARMI   ENGINEERING   dancing
False     SMIT    MECHANICAL   singing
True    HEMISH  DATA SCIENCE    gaming
False  DIVYESH           BCA    sports
True    URVISH          MBBS  swimming
IQRA S021
```